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Research Briefing | UK

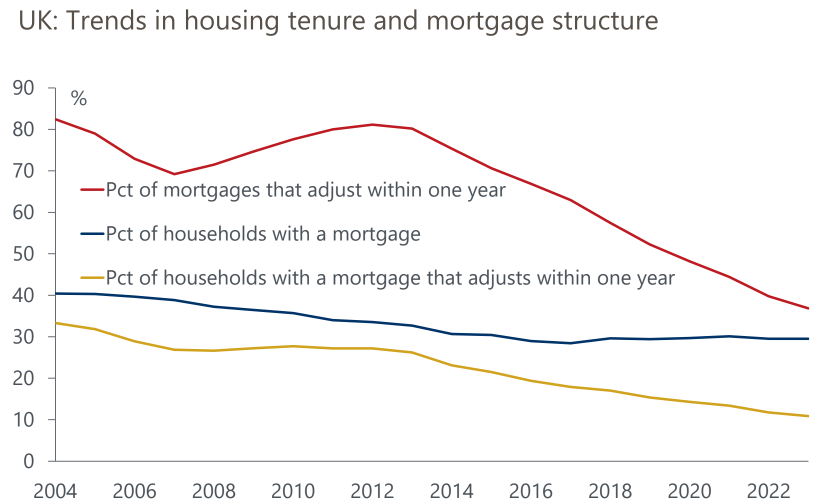
Monetary policy transmission has changed, but remains powerful

- **Over the last 10-20 years, the UK has seen a marked drop in the share of households with a mortgage, a large shift to 5-year fixed rate mortgages rather than variable rate loans, and a higher level of household savings, especially among homeowners. These changes have probably reduced the speed and scale with which changes in monetary policy affect demand through the household cashflow channel.**
- **If, like 10-20 years ago, the UK still had a high share of households with a mortgage, most mortgages were floating rate, and on average mortgagors did not have much liquid assets, then the economy would probably have slowed much more by now. Indeed, we would probably be in recession.**
- **These changes do not mean that monetary policy is ineffective. The household cashflow channel will continue to gradually affect the economy as more households refinance their mortgage at higher rates. Moreover, the household cashflow channel is not the only route by which monetary policy affects the economy. Recent monetary tightening is also dampening activity through a cashflow effect on indebted companies, higher interest rates on new loans, the rising pound, and effects on other asset prices.**
- **The current tightening cycle is probably near its end. But, with relatively high pay growth and sticky services inflation, the Bank of England (BoE) will probably be much slower to cut rates than the US Federal Reserve and European Central Bank (ECB) over the next 12-18 months.**

Historically, one of the main routes whereby monetary policy works in the UK is through the household cashflow channel. Higher debt service payments constrain the finances of indebted households, forcing them to cut spending, especially if they are credit constrained. Of course, higher interest rates also lift savers' incomes. However, savers typically do not adjust their spending much to changes in interest rates, whereas debtors do. In other words, the marginal propensity to consume out of income is higher for debtors than savers. As a result, this shift of spending power from debtors to savers reduces aggregate demand.

In this note, we discuss how the cashflow channel has changed over recent years, and what this means for the overall transmission of monetary policy and the economic outlook for the UK.

Chart 1: Only 11% of households have a mortgage that fully adjusts within a year to current interest rates



Source: Oxford Economics/Haver Analytics

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Who gains and who loses from the cashflow channel?

In the UK, most household debt is mortgage debt. Among people with mortgages, their debts typically exceed their financial assets by a wide margin. Conversely, people that own their own home with no mortgage tend to have relatively high levels of financial assets and little debt. On average, renters have low levels of financial assets and debts. The cashflow channel chiefly works by squeezing the cashflow of people with mortgages and lifting the incomes of people with high savings (who tend to also own their own home with no mortgage).

The cashflow channel for monetary policy used to be relatively powerful and quick in the UK. The share of households with a mortgage was relatively high, most mortgages were floating rate, and there was a large difference between the marginal propensity to consume of debtors and savers.

Why has it changed?

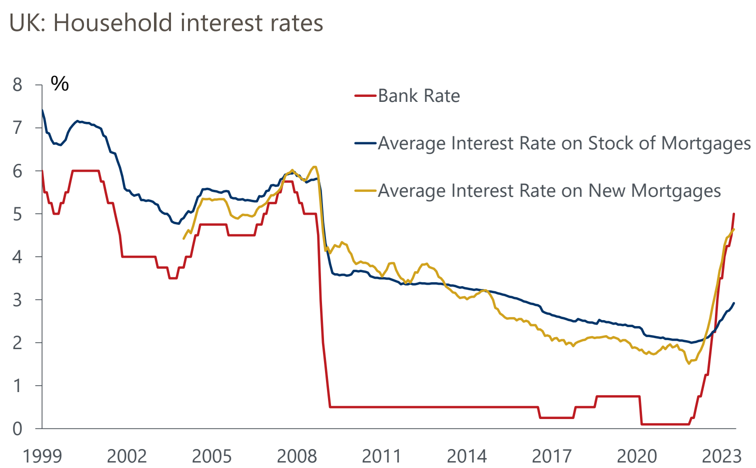
Over the last 10-20 years, there have been three big changes in household balance sheets that have affected the cashflow channel.

Fewer households have mortgages. The share of households that own their home with a mortgage in England has fallen from 42% in 2001 and 39% in 2007 to just below 30% in 2021/2022. There have been corresponding increases in the share that own their home outright (from 25% in 1995 to 35% in 2021/2022), with little change in the share that rent (but there has been shift from local council or social housing to renting privately). As a result, the aggregate household debt/income ratio has fallen from 154% in Q1 2008 to 125% in Q1 this year, with mortgage debt down from 125% to 100%.

Most mortgages used to be floating rate, and now tend to be 5-year fixed rates. In Q2 2013, 70% of outstanding mortgages were variable rate, roughly 15% were 2-year fixed, 4% were fixed for 3-4 years, and about 11% were fixed for 5 years or more. So, at that time, more than 80% of existing mortgages would fully adjust to current interest rates within a year. Now, only 13% of outstanding mortgages are variable rate, 22% are 2-year fixed, 5% are fixed for 3-4 years, 57% are 5-year fixed, and 4% are fixed for over 5 years. Less than 40% of mortgages fully adjust to current interest rates in a year. In just a few years, UK mortgage debt has shifted from being very responsive to changes in interest rates to being mostly fixed for five or more years. Therefore, changes in interest rates affect mortgages only gradually.

As a result, passthrough of the recent interest rate rises to outstanding mortgages has been limited so far. From late-2021 to June this year, the BoE policy rate increased by 490bps, with similar rises in 2- and 5-year swap rates. The passthrough to new mortgage rates was similar to historical norms: variable mortgage rates rose roughly in line with Bank Rate and fixed rates rose roughly in line with swap rates. However, the average rate on all outstanding mortgages rose by only 90bps (from 2.0% in November 2021 to 2.92% in June this year), with the average interest rate on outstanding fixed mortgages up just 50bps (at 2.4-2,5%).

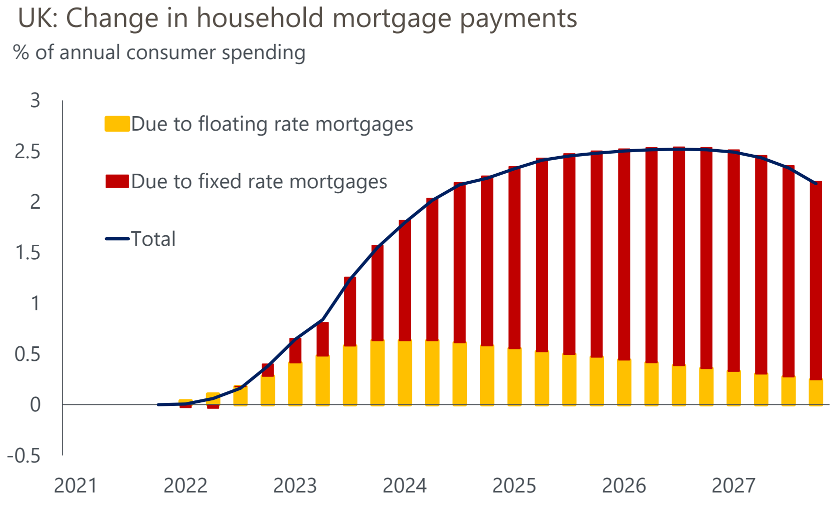
Chart 2. Average interest rate on outstanding mortgages has risen only slightly so far



Source: Oxford Economics/Haver Analytics

The current OE forecasts project that Bank Rate will rise to 5.75% later this year, then fall to 5.25% at end-2024, 4.25% at end-2025, and 2.25% at end-2027. Using this, and assuming this trajectory is reflected in swap rates and with normal spreads on new mortgages, the average interest rate on all outstanding mortgage debt will rise by a total of about 250bps from the late-2021 level (i.e. to around 4.5% in 2025-2026), with an average rise of about 30bps per quarter until mid-2024, a slower rise to late-2026, and a slight drop thereafter. The total rise in mortgage payments by 2025-2026 will equal about 2.5% of annual consumer spending, but only about one-third of this has occurred so far. By contrast, if the maturity of mortgage debt was the same as in 2013, then the average mortgage rate on existing mortgages would already have risen by about 275bps so far, with a further rise likely in the near term and a much steeper fall in 2025-2027.

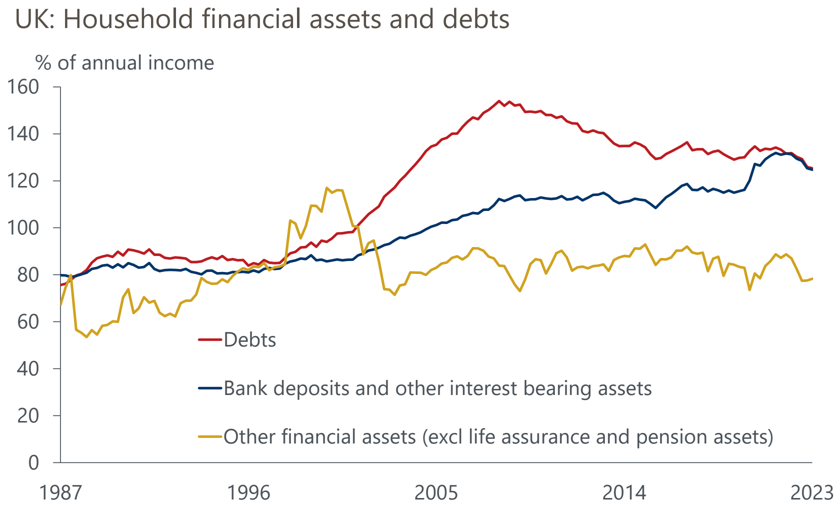
Chart 3: Most of the rise in mortgage payments on outstanding loans is yet to come



Source: Oxford Economics/Haver Analytics

Higher level of savings relative to debts. Since Q1 2008, households' holdings of bank deposits and other interest-bearing assets have risen from 112% of income to 125%, and now roughly equal household debts. Including holdings of other financial assets (e.g. equities, unit trusts, life assurance, and pension fund assets), total household net financial wealth rose from 250% of annual income mid-2008 to 328% in Q1 this year. Microdata from the BoE NMG survey suggest that average holdings of financial assets rose markedly between 2014 and 2020 among both homeowners and mortgagors, with little change among renters.

Chart 4: Household debts have fallen relative to incomes, while savings have risen



Source: Oxford Economics/Haver Analytics

Note that the shift to long term fixed rate products has not been replicated for household savings. The share of household bank deposits that are in time deposits has risen recently but, at 31% in June, remains lower than 10 years earlier (44%). Moreover, most household time deposits have an original maturity below two years, and as a result a relatively large share have adjusted to recent higher interest rates. For example, the average interest rate on households' time deposits has risen by 240bps since November 2001, far more than the average 90bps rise in outstanding mortgages. The result is that household interest receipts have risen much faster than interest payments.

These trends (lower home ownership, less floating rate debt, and higher savings) are not unique to the UK but have been relatively pronounced here. For example, since 2007, the share of households with a mortgage in the UK has fallen by 9ppts, a bigger drop than in any EU country. Similarly, the floating rate share of new mortgages has fallen further in the UK, dropping by 44ppts from 52% during 2004-2010 to 8% since the start of 2017. The same figure for the eurozone has fallen by 27ppt from 45% to 18%. Since 2017, the share of new mortgages with variable rates in the UK has been among the lowest in Europe.

How has this affected the transmission of monetary policy?

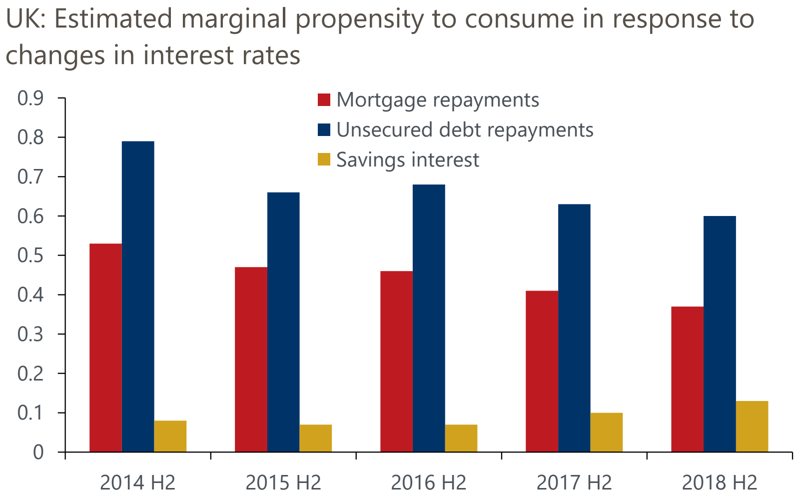
These changes have caused the cashflow channel of monetary policy to change. It has become slower, more narrowly focussed on a smaller pool of households, and probably less effective overall.

The slower pass through to interest rates on outstanding mortgages is likely to result in a more gradual adjustment in spending. Even if you assume households are a bit forward-looking, it is unlikely that someone currently on a 5-year fixed mortgage that does not need to refinance until 2026 or 2027 will cut spending a lot in anticipation of higher mortgage payments three-four years ahead. This is supported by research from Sweden (where the share of variable rate mortgages is much higher than in the UK), which suggests that, in response to changes in interest rates, households with a variable rate mortgage adjust their spending much more sharply than households with a fixed rate mortgage.

Given the decline in the household debt/income ratio (and the share of households with a mortgage), as well as the rise in households' interest-bearing assets, a given change in interest rates deducts less from household income in higher interest payments and adds more in higher interest receipts than it did before. This by itself would weaken the cashflow channel.

This effect has probably been reinforced by a decline in the marginal propensity to consume (mpc) among debtors in response to changes in interest rates. Results from the BoE NMG survey suggest that the mpc in response to changes in mortgage payments fell from 0.53 in 2014 to 0.37 in 2018 (so that if mortgage bills rise by £100, then mortgagors cut spending by £37), while the mpc for unsecured debt fell from 0.79 to 0.60 over the same period. By contrast, the mpc for changes in interest income rose from 0.08 to 0.13 between 2014 and 2018.

Chart 5: Spending among people with debts has become less sensitive to changes in interest rates



Source: Oxford Economics / Haver Analytics

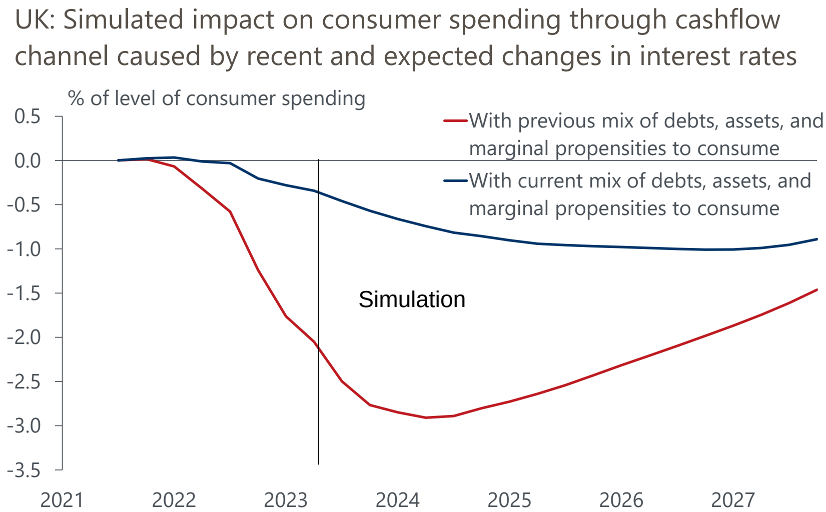
The BoE research attributes this decline in mortgagors' mpc to the relatively easy availability of credit and the marked rise in their average holdings of liquid assets, which provide a financial buffer that allows people to avoid having to adjust spending one-for-one in response to higher interest outlays. This is consistent with cross-section research which shows that among mortgagors, the mpc in response to changes in interest rates is typically lower among people with higher levels of liquid assets. Our analysis of the NMG microdata suggests that the mpc among mortgagors fell further in 2019 and 2020.

We can combine these effects into a simple overall model of the household cashflow channel. In this simple model, we use the actual changes in mortgage rates and savings rates up to June (latest data) and assume (as before) that interest rates evolve in line with OE forecasts.

In the first simulation, we use the current levels of debt/income and assets/income, the current split of the maturity of mortgages, and the BoE's latest estimates for mpcs for interest income and payments (in 2018). In the second, we assume that the ratios of household debt/income and assets/income, as well as the maturity structure of mortgage debt, equal the 2004-2013 averages, and that the mpcs for interest receipts and payments equal the BoE estimates for 2014.

The difference between these two simulations is stark. In the first, the cashflow effect cuts about 0.3ppts off consumer spending in Q2 this year, with a slightly greater drag in Q4, but will cut spending by nearly 1% in Q4 next year and around 1% in 2025-2026, with only a slightly smaller drag in 2027. So even with lower interest rates in 2024-2026, the cashflow channel in that period will not provide any fresh support to consumer spending because of the continued refinancing of expiring fixed rate loans. In the second, the cashflow effect would have cut about 2% off the level of consumer spending in Q2, building to a drag of about 3% in Q4 this year, and creating a similar drag next year and a lower drag thereafter. The economy would probably have slowed much more by now. Indeed, we would probably be in recession. To stress, both scenarios use the same inputs for interest rates on new loans. The differences stem from the lower debt/income ratio, higher ratio of assets/income, slower adjustment in mortgage payments, and smaller gaps between the mpc on interest payments and receipts.

Chart 6: The cashflow channel on consumer spending has become slower and less powerful



Source : Oxford Economics/Haver Analytics

These results are consistent with BoE research, which suggests that if the cashflow effect is restricted to savings and debts that are variable rate or which adjust within two years, then the near-term cashflow effect on consumer spending is likely to be small. It is only later that household borrowing costs adjust enough that cashflow effects begin to work to restrain demand.

The slower impact of monetary policy could further reduce its effect on inflation. For example, if inflation is lifted above target (e.g. due to a demand shock), then higher interest rates would now take longer to return inflation to target. If inflation expectations are partly adaptive, this would make it more likely that inflation expectations rise above target-consistent levels, which would make it more likely that second-round effects on pay growth keep inflation above target. Ultimately, this would probably mean that a bigger total rise in interest rates (or a longer period of high rates) is needed to return inflation to target. Something like this seems to be playing out at present.

As well as being slower, the impact of monetary policy is much narrower. So far, only about a quarter of existing mortgagors (and hence only 7%-8% of all households) have faced the higher mortgage rates charged this year. Most households haven't yet felt this squeeze directly, either because they don’t have a mortgage and or because their mortgage fixed before this year. But individual households who refinance a maturing loan (or take out a new mortgage) now face a very sharp jump in their mortgage payments. The average rate on expiring fixed rate mortgages is around 2%, and current mortgage rates are around 5.75%. For someone with a £200,000 mortgage (the average approval for refinancing an existing mortgage was £205,000 over the last year), this would lift monthly mortgage payments by about £400 (using a standard mortgage repayment calculator program), about 8% of average disposable income among households with a mortgage in 2021/2022 and a higher share of income for the average household. Each month, roughly 2% of fixed rate mortgages (i.e. 0.6%-0.7% of households) need to be refinanced, and for these households the jump in mortgage payments is likely to be brutally painful.

One could argue that such a severe cashflow impact might raise the mpc among mortgagors, because they would lack sufficient savings to smooth spending. However, in practice, other behavioural or policy changes may offset this. Concern that households will be unable to cope with the mortgage refinancing crunch has led to an agreement between lenders, the FCA, and the government permitting customers to switch to an interest-only mortgage for six months, or extend their mortgage term to reduce their monthly payments and switch back to their original term within the first six months, if they choose to. For someone with a £200,000 mortgage, a shift to an interest-only mortgage would cut repayments by about £270. A shift from a 25-year to 35-year mortgage would cut monthly payments by about £150 (although this may not be available for many people, given difficulties in extending a mortgage beyond expected retirement). Such forbearance is understandable and does not pose a risk to financial stability. The average homeowner re-mortgaging over the last twelve months had around a 50% loan-to-value ratio, implying considerable scope for lenders to postpone further repayment of principal. But the more households opt for such forbearance, the more the mpc among mortgagors is reduced and the ability of monetary policy to restrain demand is diluted.

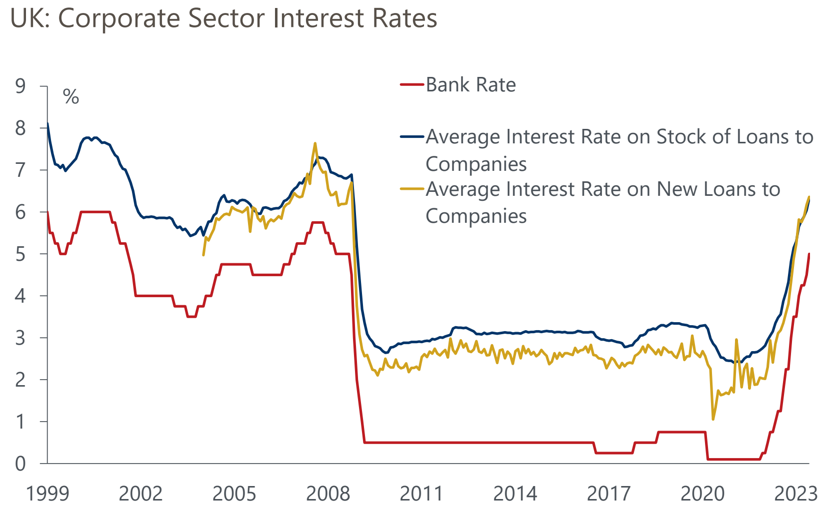
'Less effective' does not mean 'ineffective'

These changes to the cashflow channel imply that the overall effect of monetary policy on demand is probably somewhat slower and smaller than before. This does not mean that monetary policy has become totally ineffective. The share of fixed rate mortgages has long been relatively high in the US, France, and Germany, implying a less powerful effect from the cashflow channel. Even so, few doubt that monetary policy still works in those countries.

The household cashflow channel is only part of the monetary transmission process. For example, higher interest rates also squeeze the finances of indebted companies, a corporate version of the cashflow channel. Roughly 80% of UK companies' sterling-denominated bank debt is floating rate, and this share has not changed much in recent years. The average interest rate on UK companies' outstanding bank debt has risen by 330bps since late-2021, a far greater rise than for outstanding mortgages.

With profits also under pressure, corporate cashflow is weak. The sterling bank deposits of UK private non-financial companies are down 4.6% y/y, the sharpest drop since 1980. In prior cycles, sharp declines in companies' bank deposits usually have been a precursor to broader cutbacks in jobs and capex.

Chart 7: Unlike households, interest rates on outstanding corporate bank debt have risen sharply



Source: Oxford Economics/Haver Analytics

Changes in interest rates also affect demand through inter-temporal substitution, whereby higher interest rates on new loans discourage households and firms from borrowing to spend and lead to higher savings. Interest rate changes also affect the economy through the exchange rate and other asset prices (e.g. equity prices, credit spreads). There are signs that the sharp tightening in monetary policy is feeding through these other channels. For example, housing is weak, and credit growth has slowed markedly. Firms' investment intentions are soft, and the share of firms in both services and manufacturing who report that investment is constrained by the cost of finance has risen sharply. With rising differentials in the forward path of interest rates, sterling's trade weighted exchange rate has risen by about 6% since the start of the year, which will reduce import prices (reducing inflation directly) but also dampen growth through a deterioration in net trade.

The overall effect is that the channels through which monetary policy affects the economy may be less centred on consumer spending than previously. Instead, they are more reliant on the financial position of companies, net trade, investment, and employment.

The speed and extent to which monetary policy affects the economy through these other channels probably is more uncertain and variable than the rather quick, mechanical, and predictable effects that the cashflow channel used to provide. For example, the impact of higher interest rates on credit spreads and asset prices occurs partly through a higher discount rate for future cashflows. But it also partly comes about through the role of higher interest rates in slowing the economy through other channels, thereby reducing expectations for incomes and profits, and lifting unemployment and insolvencies. These indirect effects may be slow to come through. But this implies that an economic slowdown, once it occurs, might well be reinforced by tighter financial conditions through wider credit spreads and weaker asset prices.

Slower-acting monetary policy makes it harder to fine tune the economy

If the monetary transmission process is slower, somewhat less effective, and less predictable, then this would also imply reduced scope for the BoE to finetune policy in a way that leaves the economy with a zero output gap and inflation on target. This would tend to reinforce the prospect that both growth and inflation in the coming years will be more volatile than during the 25 years before the pandemic.

The BoE could (to an extent) have compensated for this with an earlier and larger initial policy response to achieve the desired rapid restraint on demand and inflation. But once a substantial amount of tightening already is in place (like now), the BoE are likely to be aware that even if the policy rate and interest rates on new loans stop rising, the economy will face a powerful delayed monetary tightening through the refinancing of expiring fixed rate mortgages over the next few years. This is likely to dampen spending for an extended period. As a result, the BoE are likely to be reluctant to hike rates much further and will prefer to transition to a strategy of keeping rates on hold while the lagged effects of previous tightening work through the economy. Indeed, as our simulation shows, even if the BoE policy rate were to fall slightly next year, the average interest rate on the stock of mortgage debt will probably continue to rise as existing mortgages with rates around 2% are refinanced at higher rates.